

July 2025 CSE 209

Computer Architecture

Arithmetic Logic Unit (ALU)

Arithmetic Logic Unit (ALU)

- Can perform both Arithmetic and Logic Operations

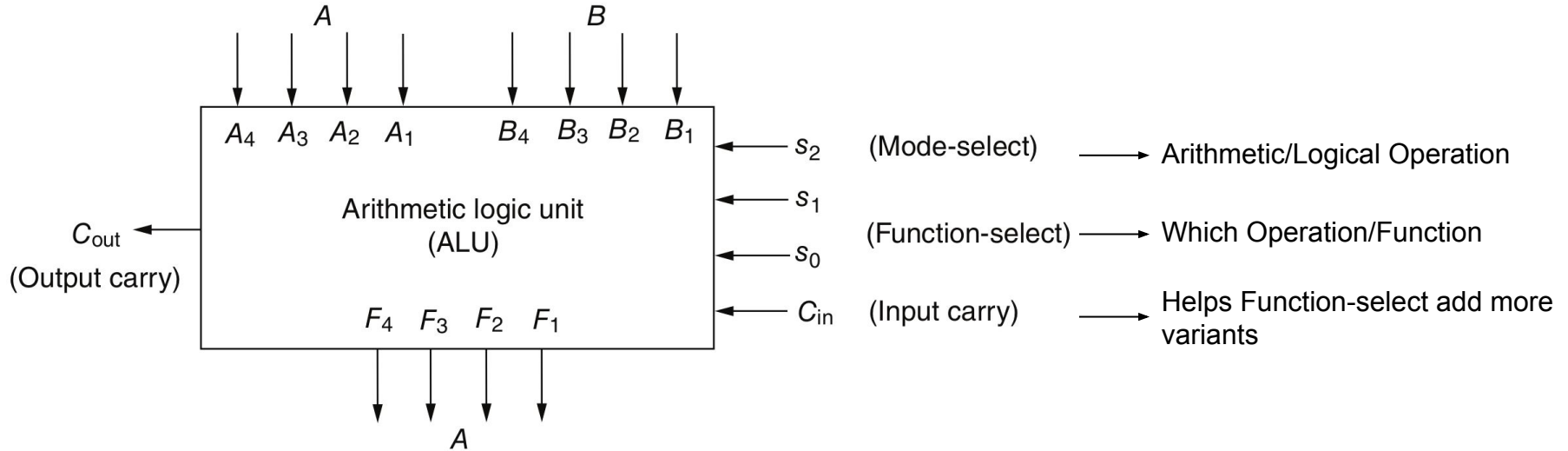
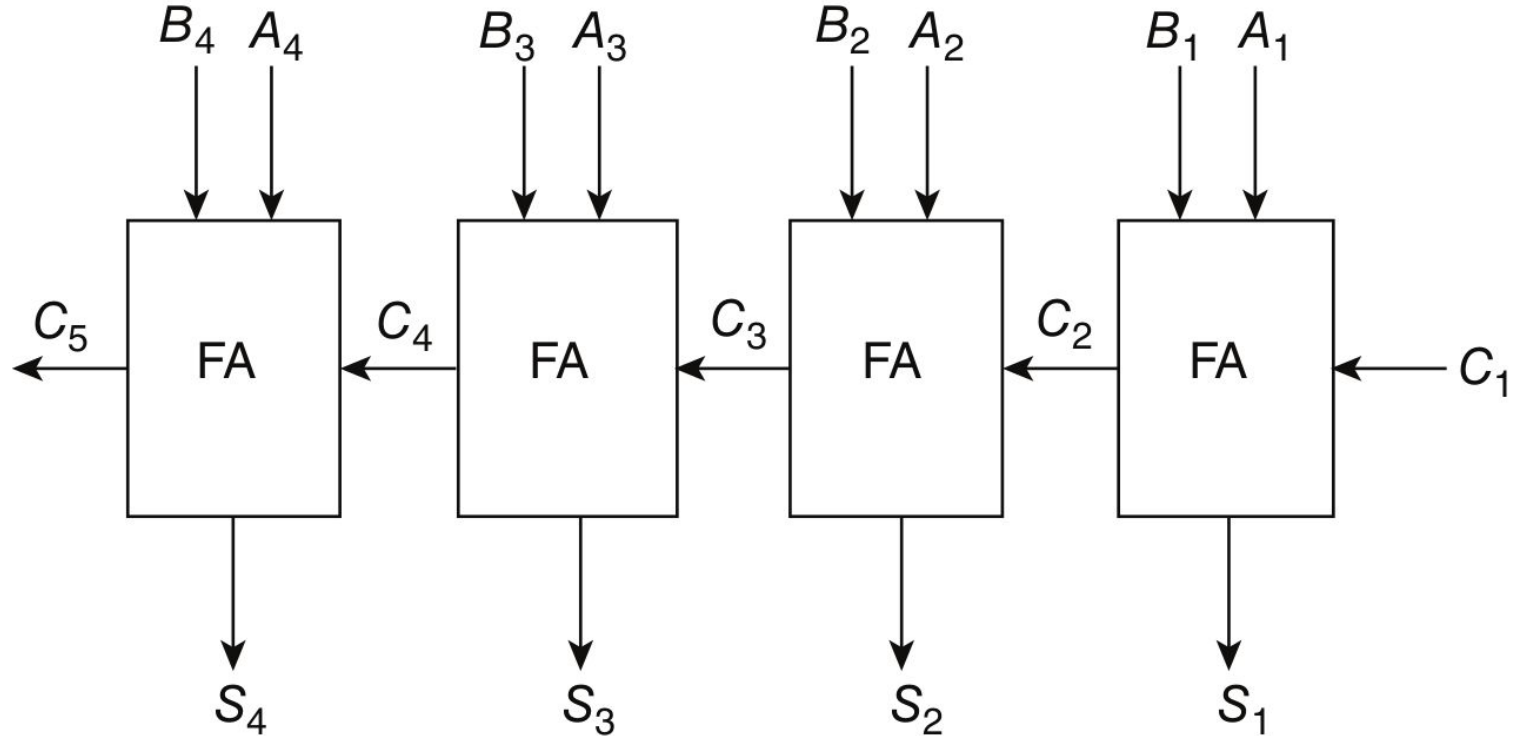


Figure 9-5 Block diagram of a 4-bit ALU

Design of Arithmetic Circuit

- Parallel adder
 - A number of full-adder circuits connected in cascade

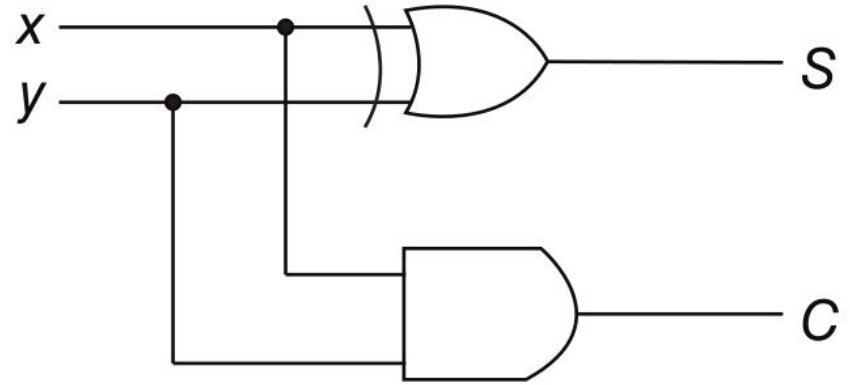


1-bit Half-Adder

x	y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

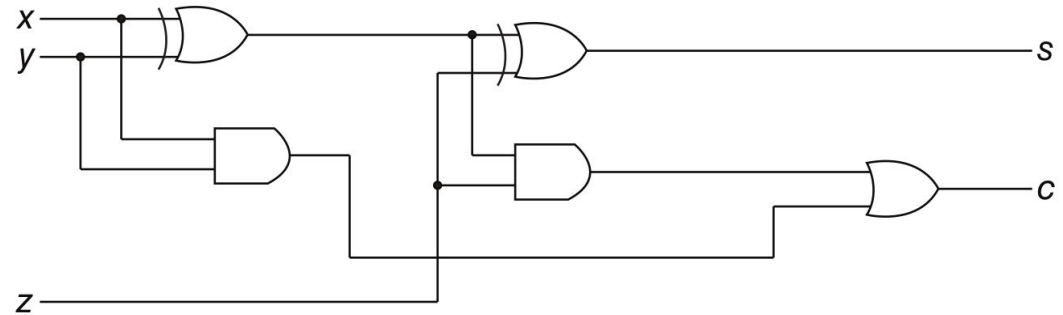
$$S = x'y + xy'$$

$$C = xy$$



1-bit Full-Adder

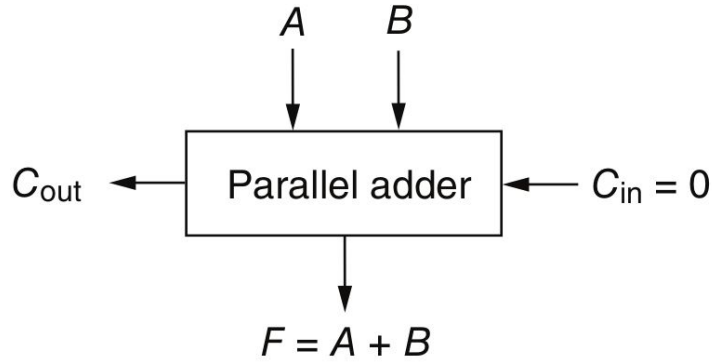
x	y	z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



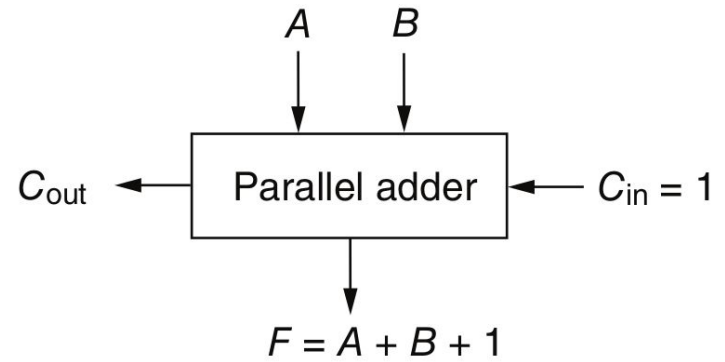
$$S = x'y'z + x'yz' + xy'z' + xyz$$

$$C = xy + xz + yz$$

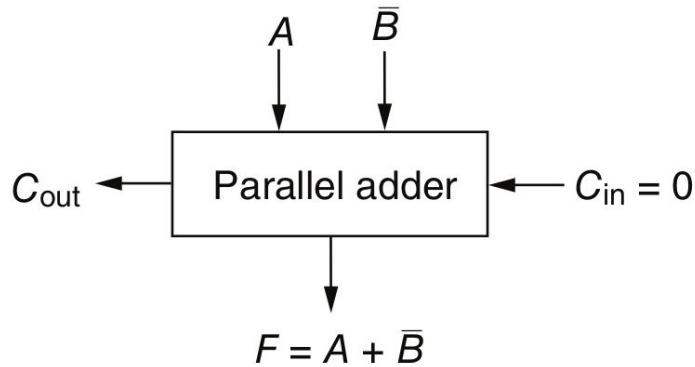
Arithmetic Operations by ALU



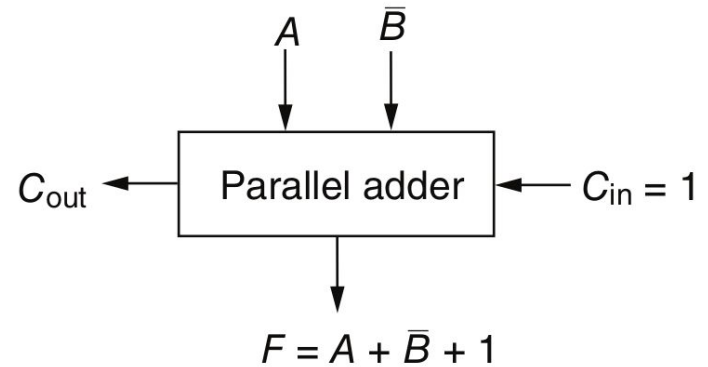
(a) Addition



(b) Addition with carry

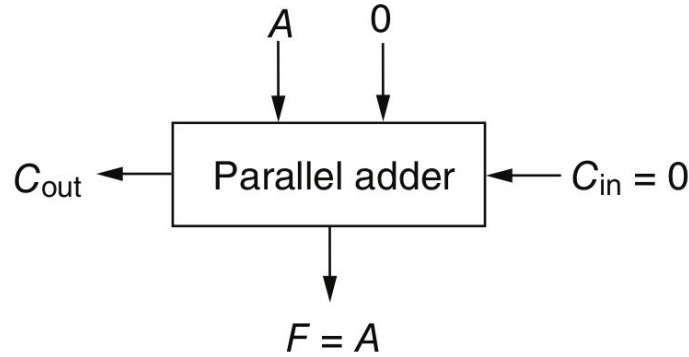


(c) A plus 1's complement of B

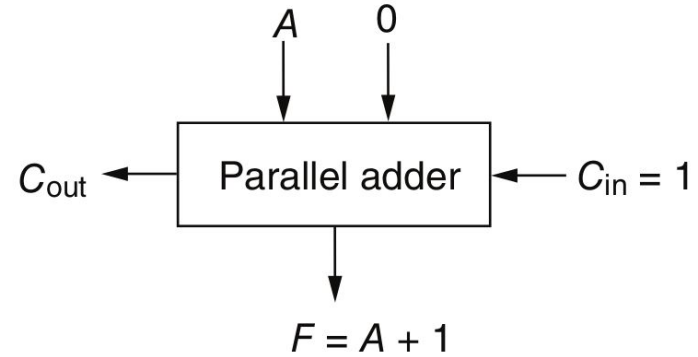


(d) Subtraction

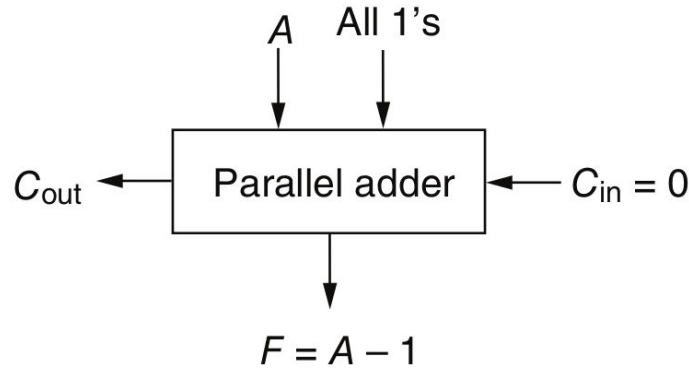
Arithmetic Operations by ALU



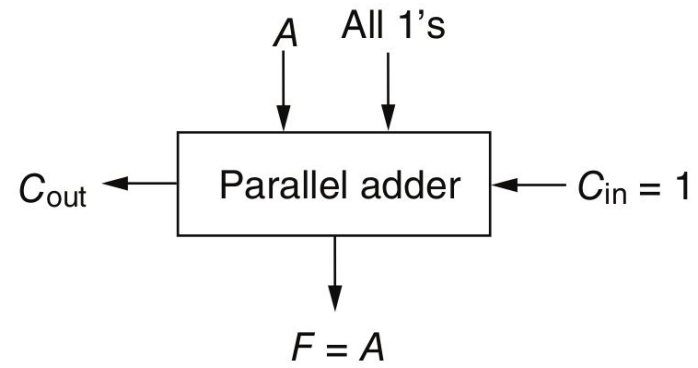
(e) Transfer A



(f) Increment A

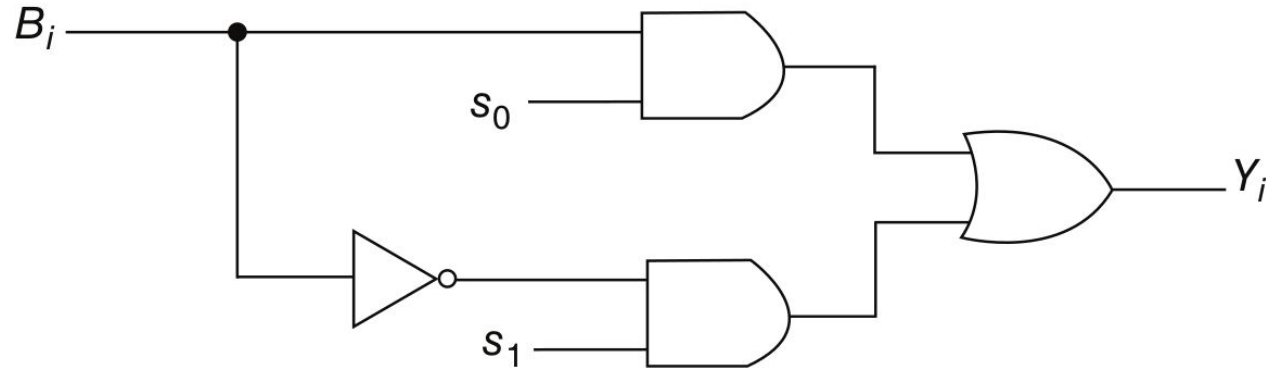


(g) Decrement A



(h) Transfer A

Arithmetic Operations by ALU



s_1	s_0	Y_i
0	0	0
0	1	B_i
1	0	B'_i
1	1	1

Figure 9-7 True/complement, one/zero circuit

$$X_i = A_i$$

$$Y_i = B_i s_0 + B'_i s_i \quad i = 1, 2, \dots, n$$

Arithmetic Operations by ALU

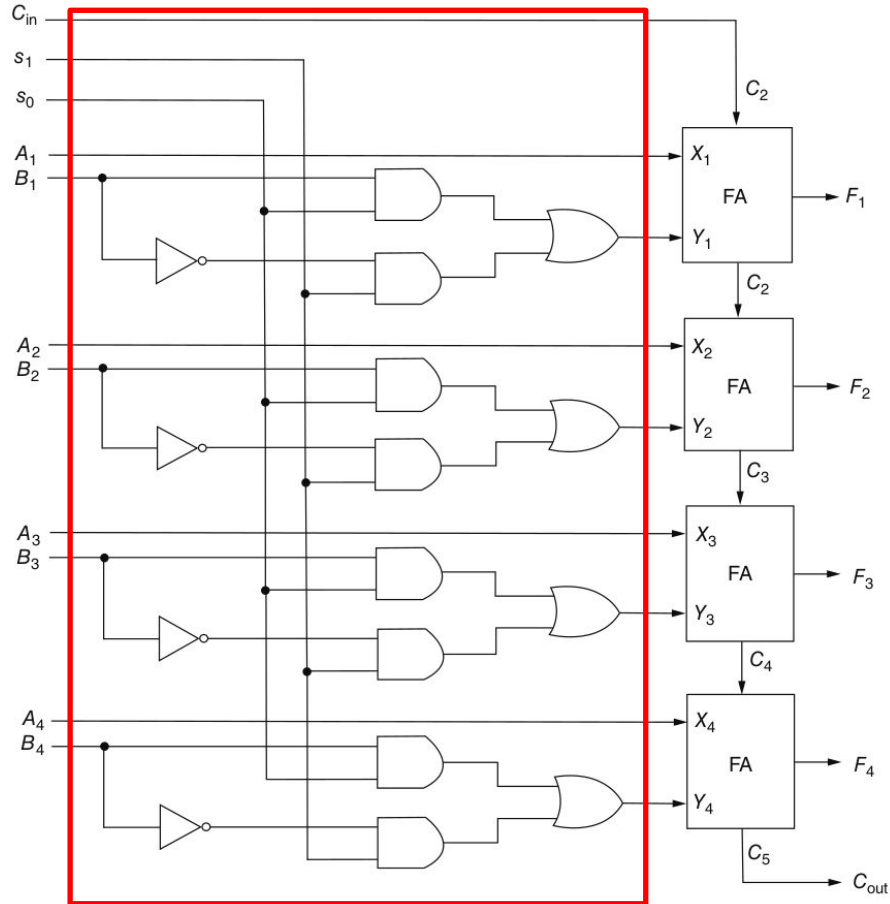
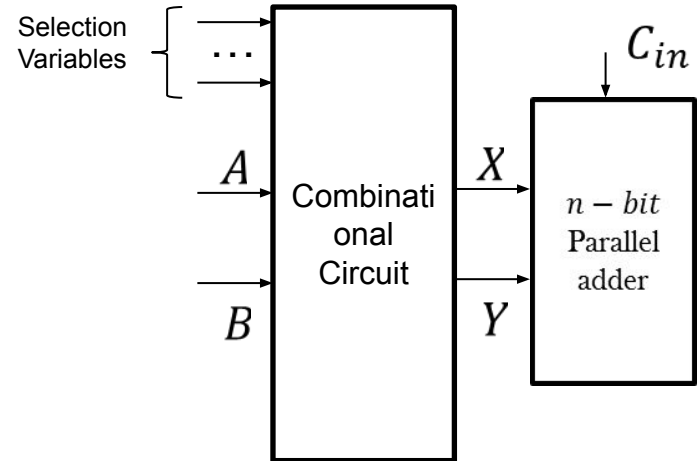


Figure 9-8 Logic diagram of arithmetic circuit



Arithmetic Operations by ALU

Table 9-1 Function table for the arithmetic circuit of Fig. 9-8

Function select			Y equals	Output equals	Function
s_1	s_0	C_{in}			
0	0	0	0	$F = A$	Transfer A
0	0	1	0	$F = A + 1$	Increment A
0	1	0	B	$F = A + B$	Add B to A
0	1	1	B	$F = A + B + 1$	Add B to A plus 1
1	0	0	$\bar{B}$	$F = A + \bar{B}$	Add 1's complement of B to A
1	0	1	$\bar{B}$	$F = A + \bar{B} + 1$	Add 2's complement of B to A
1	1	0	All 1's	$F = A - 1$	Decrement A
1	1	1	All 1's	$F = A$	Transfer A

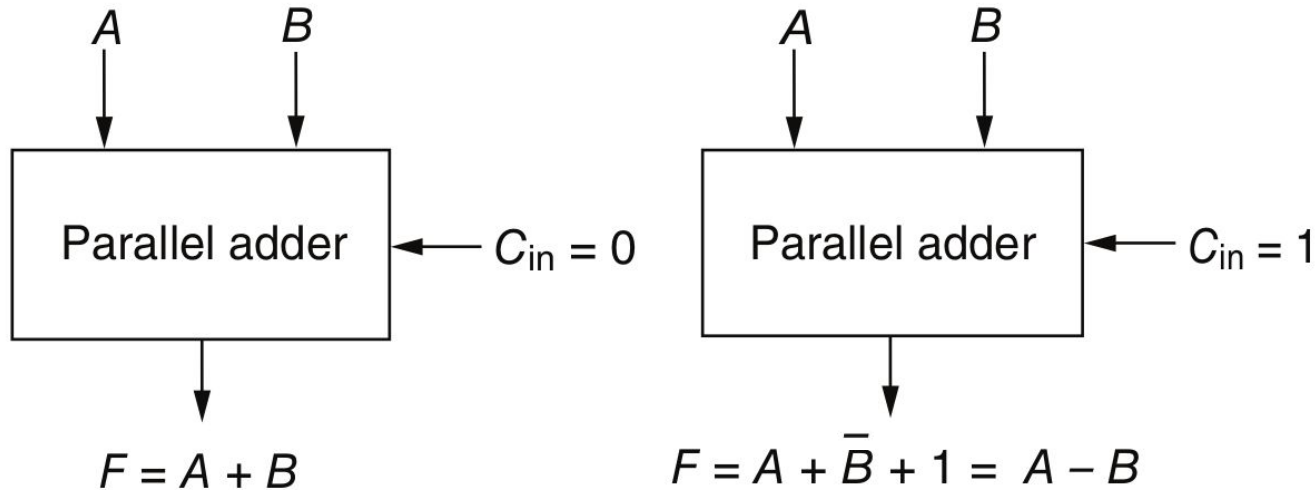
Arithmetic Operations by ALU

Table 9-2 Effect of output carry in the arithmetic circuit of Fig. 9-8

Function select			Arithmetic function	$C_{\text{out}} = 1$ if	Comments
s_1	s_0	C_{in}			
0	0	0	$F = A$		C_{out} is always 0
0	0	1	$F = A + 1$	$A = 2^n - 1$	$C_{\text{out}} = 1$ and $F = 0$ if $A = 2^n - 1$
0	1	0	$F = A + B$	$(A + B) \geq 2^n$	Overflow occurs if $C_{\text{out}} = 1$
0	1	1	$F = A + B + 1$	$(A + B) \geq (2^n - 1)$	Overflow occurs if $C_{\text{out}} = 1$
1	0	0	$F = A - B - 1$	$A > B$	If $C_{\text{out}} = 0$, then $A < B$ and $F = 1$'s complement of $(B - A)$
1	0	1	$F = A - B$	$A \geq B$	If $C_{\text{out}} = 0$, then $A < B$ and $F = 2$'s complement of $(B - A)$
1	1	0	$F = A - 1$	$A \neq 0$	$C_{\text{out}} = 1$, except when $A = 0$
1	1	1	$F = A$		C_{out} is always 1

Design Example

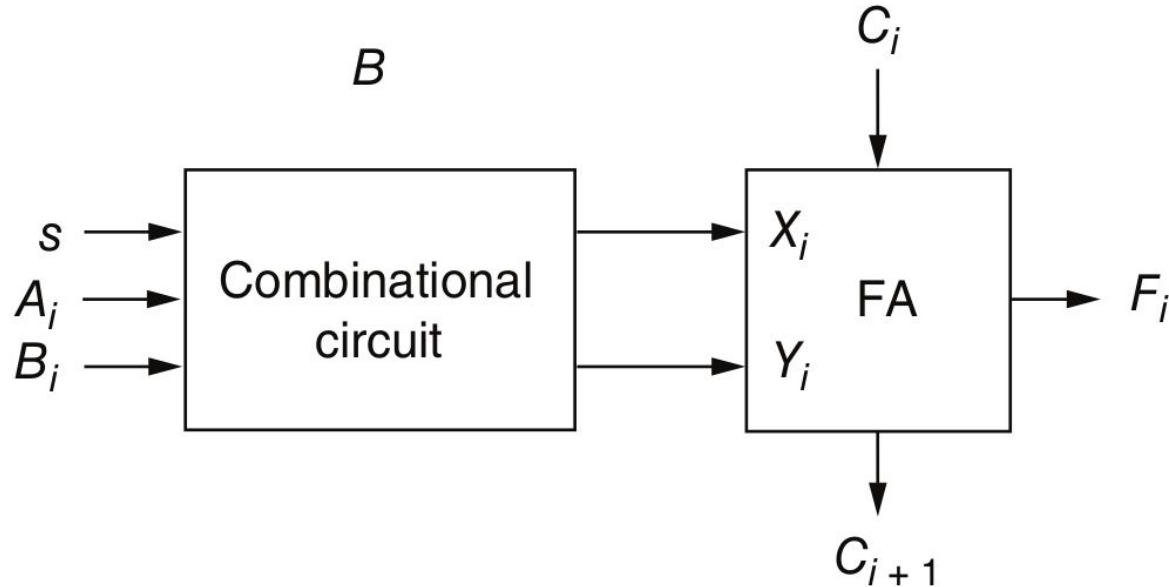
Design an adder/subtractor circuit with one selection variable s and two inputs A and B . When $s = 0$ the circuit performs $A + B$. When $s = 1$ the circuit performs $A - B$ by taking the 2's complement of B .



(a) Function specification

Design Example

Design an adder/subtractor circuit with one selection variable s and two inputs A and B . When $s = 0$ the circuit performs $A + B$. When $s = 1$ the circuit performs $A - B$ by taking the 2's complement of B .



Design Example

Design an adder/subtractor circuit with one selection variable s and two inputs A and B . When $s = 0$ the circuit performs $A + B$. When $s = 1$ the circuit performs $A - B$ by taking the 2's complement of B .

s	A_i	B_i	X_i	Y_i
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	0
1	1	0	1	1
1	1	1	1	0

$$X_i = A_i$$

$$Y_i = B_i \oplus s$$

$$C_{in} = s$$

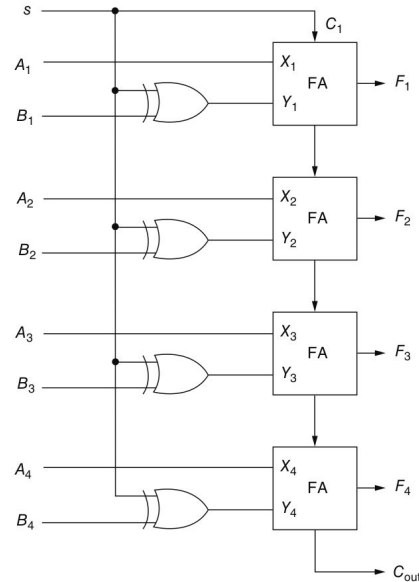
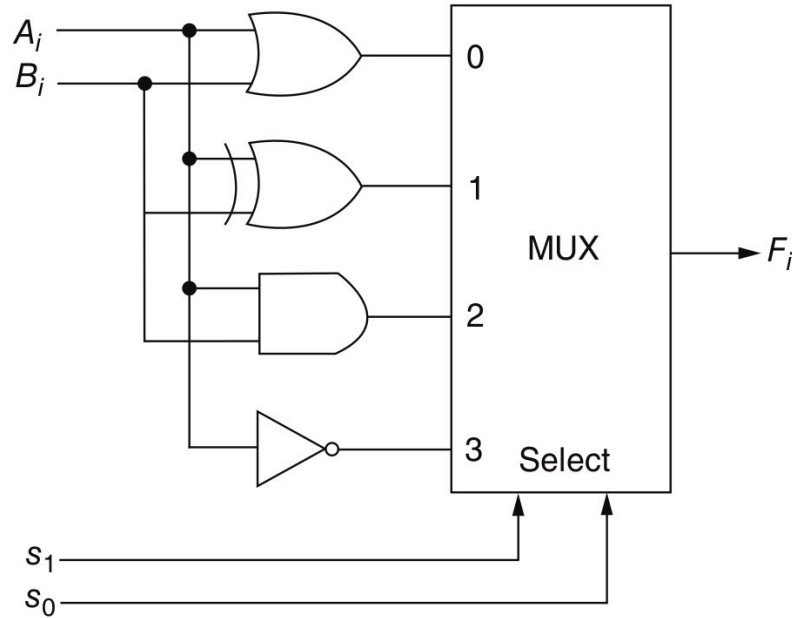


Figure 9-10 4-bit adder/subtractor circuit

(c) Truth table and simplified equations

Design of Logic Circuit



(a) Logic diagram

s_1	s_0	Output	Operation
0	0	$F_i = A_i + B_i$	OR
0	1	$F_i = A_i \oplus B_i$	XOR
1	0	$F_i = A_i B_i$	AND
1	1	$F_i = A_i'$	NOT

(b) Function table

Figure 9-11 One stage of logic circuit

Design of Logic Circuit

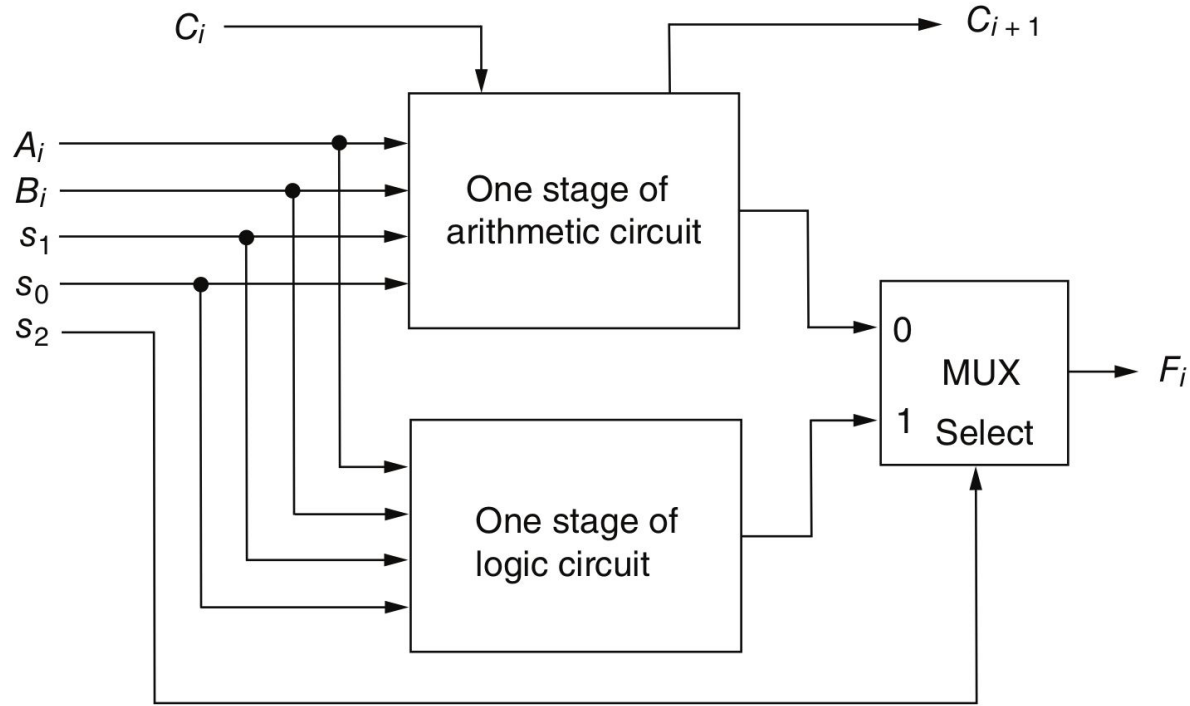


Figure 9-12 Combining logic and arithmetic circuits

More Efficient Design

1. Design the arithmetic section independent of the logic section.
2. Determine the logic operations obtained from the arithmetic circuit in step 1, assuming that the input carries to all stages are 0.
3. Modify the arithmetic circuit to obtain the required logic operations.

Table 9-3 Logic operations in one stage of arithmetic circuit

s_2	s_1	s_0	X_i	Y_i	C_i	$F_i = X_i \oplus Y_i$	Operation	Required operation
1	0	0	A_i	0	0	$F_i = A_i$	Transfer A	OR
1	0	1	A_i	B_i	0	$F_i = A_i \oplus B_i$	XOR	XOR
1	1	0	A_i	B'_i	0	$F_i = A_i \odot B_i$	Equivalence	AND
1	1	1	A_i	1	0	$F_i = A'_i$	NOT	NOT

More Efficient Design

1. Design the arithmetic section independent of the logic section.
2. Determine the logic operations obtained from the arithmetic circuit in step 1, assuming that the input carries to all stages are 0.
3. Modify the arithmetic circuit to obtain the required logic operations.

$$X_i = A_i + s_2 s_1' s_0' B_i + s_2 s_1 s_0' B_i'$$

$$Y_i = s_0 B_i + s_1 B_i'$$

$$Z_i = s_2' C_i$$

More Efficient Design

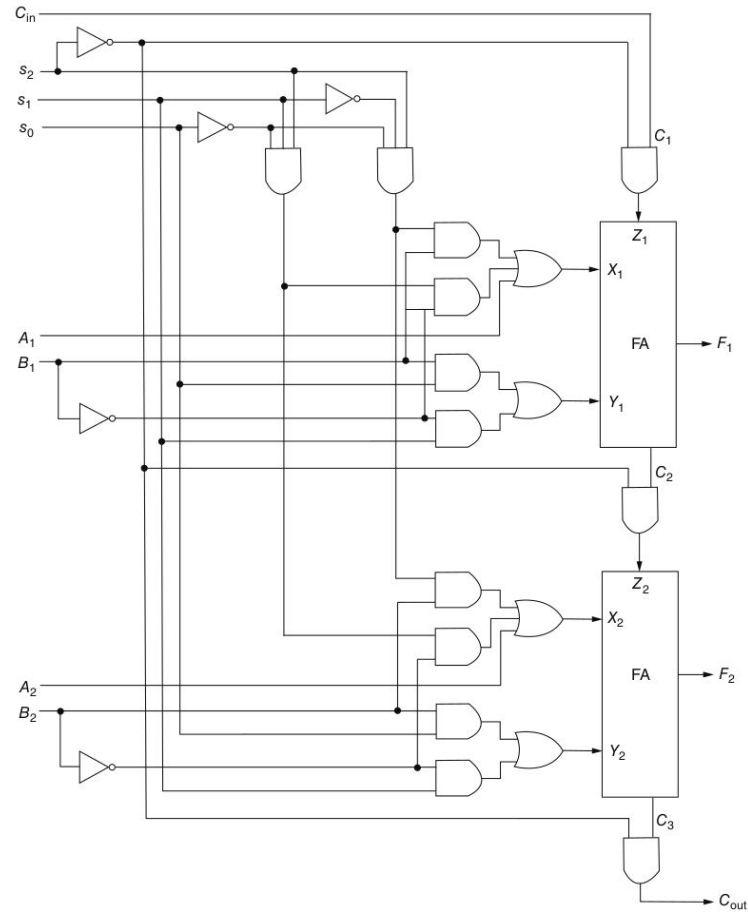


Figure 9-13 Logic diagram of arithmetic logic unit (ALU)

More Efficient Design

Table 9-4 Function table for the ALU of Fig. 9-13

Selection				Output	Function
s_2	s_1	s_0	C_{in}		
0	0	0	0	$F = A$	Transfer A
0	0	0	1	$F = A + 1$	Increment A
0	0	1	0	$F = A + B$	Addition
0	0	1	1	$F = A + B + 1$	Add with carry
0	1	0	0	$F = A - B - 1$	Subtract with borrow
0	1	0	1	$F = A - B$	Subtraction
0	1	1	0	$F = A - 1$	Decrement A
0	1	1	1	$F = A$	Transfer A
1	0	0	X	$F = A \vee B$	OR
1	0	1	X	$F = A \oplus B$	XOR
1	1	0	X	$F = A \wedge B$	AND
1	1	1	X	$F = \bar{A}$	Complement A

Status Register

1. Bit C is set if the output carry of the ALU is 1. It is cleared if the output carry is 0.
2. Bit S is set if the highest-order bit of the result in the output of the ALU (the sign bit) is 1. It is cleared if the highest-order bit is 0.
3. Bit Z is set if the output of the ALU contains all 0's, and cleared otherwise. $Z = 1$ if the result is zero, and $Z = 0$ if the result is nonzero.
4. Bit V is set if the exclusive-OR of carries C8 and C9 is 1, and cleared otherwise. This is the condition for overflow when the numbers are in sign-2's-complement representation. For an 8-bit ALU, V is set if the result is greater than 127 or less than -128.

Status Register

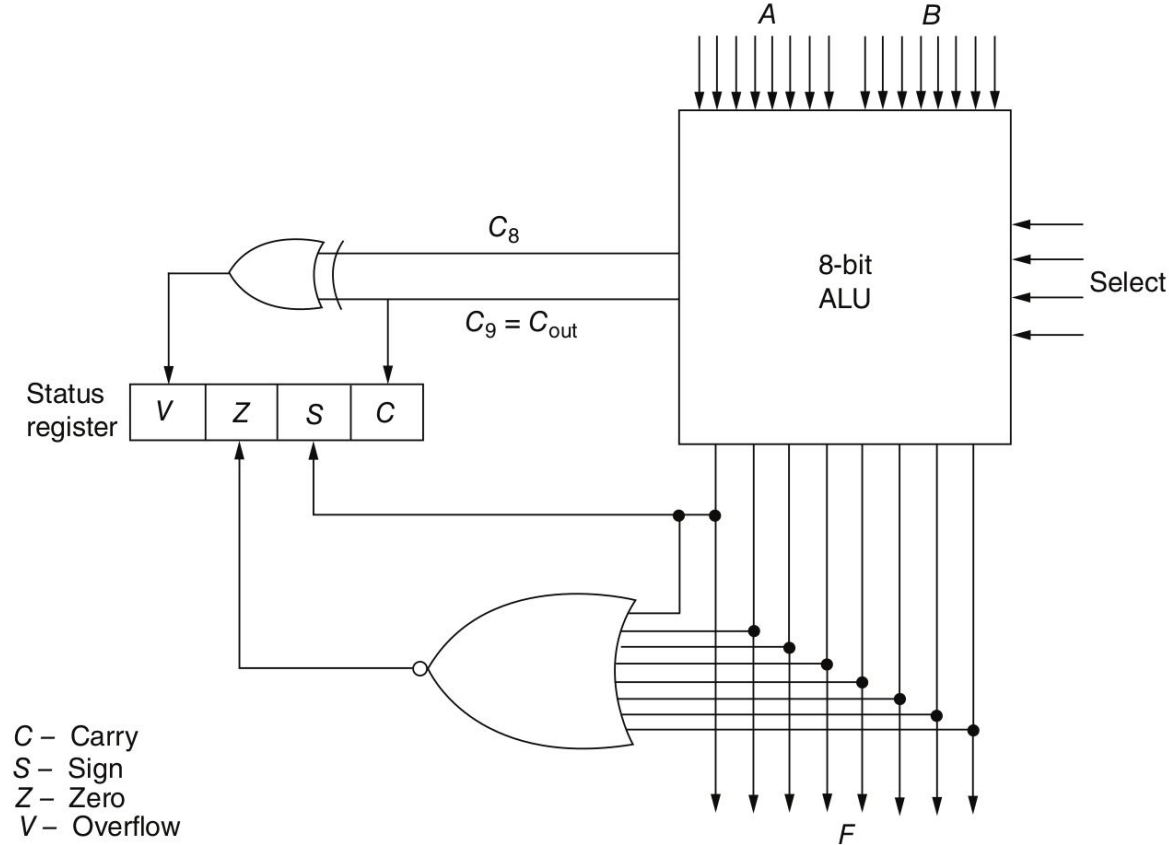


Figure 9-14 Setting bits in a status register

References

Digital Logic and Computer Design by M. Morris Mano (Chapter 9.3-9.7)